ModelIt!

Student Activity Pack

Why Does Stress Make You Sick?

Modeling Stress Hormones, Immune Suppression, and the Mind-Body Connection

Grade 12L1 | NGSS HS-LS1-3 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever gotten sick right after a really stressful period (like after finals or a big event)? Why do you think that happens?**

2. **What do you think the connection is between your mental state and your physical health? Can thoughts really affect your body?**

3. **What does 'stress' actually mean biologically? What's happening inside your body when you feel stressed?**

4. **Do you think having close friends and family actually affects your physical health, or is that just a feel-good idea?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Stress Exposure | Social Support | Cortisol Level | Immune Function | Health Outcome

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Fight or Flight

Stress Exposure: High, single event | Social Support: Moderate | Duration: 1 hour

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Finals Week

Stress Exposure: Moderate-high, continuous | Social Support: Low (isolated studying) | Duration: 2 weeks

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Supported Through Crisis

Stress Exposure: High, continuous | Social Support: High (strong friend/family network) | Duration: 2 weeks

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Sleep Quality — The duration and architecture of sleep — chronic stress disrupts sleep, and sleep deprivation independently elevates cortisol and suppresses immune function, creating a vicious cycle that compounds stress-immune damage

Inflammatory Response — The body's generalized immune activation that increases under chronic stress — cortisol paradoxically increases systemic inflammation even while suppressing specific immune defenses, contributing to autoimmune conditions and cardiovascular disease

Coping Strategy Effectiveness — The degree to which an individual's stress management techniques actually reduce HPA axis activation — evidence-based strategies (exercise, mindfulness, social connection) produce measurable cortisol reduction, while maladaptive coping (substance use, avoidance) can worsen the stress response

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Your body has an immune system powerful enough to kill cancer cells and fight off deadly pathogens. So why does being stressed about a test, a breakup, or family problems make you more likely to catch a cold, break out in acne, and take longer to heal from injuries? Use evidence from your simulation to support your answers.

1. **How does your model explain the paradox that stress evolved as a survival mechanism but now causes disease?**

2. **Why does chronic stress suppress the immune system while acute stress briefly enhances it? What does this reveal about evolutionary design?**

3. **Your model shows that social support physically reduces cortisol. What are the implications for people who are socially isolated — and for how we design schools and communities?**

4. **How might understanding the stress-immune connection change the way you manage your own health during stressful periods?**

5. **What are the equity implications of stress-immune research, given that poverty, discrimination, and adverse childhood experiences create chronic stress in specific populations?**

8 STEM Challenge: Design a School Stress Reduction Program

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YOUR ENGINEERING MISSION

Using your model data on the biological mechanisms of stress-immune suppression, design an evidence-based stress reduction program for your school that targets the specific biological pathways through which chronic stress damages health.

THE SCENARIO

Your school's health data shows that 68% of seniors report chronic stress, absences spike during exam periods, and students who report low social support have 3x more sick days. The administration wants a science-based program — not generic wellness posters — that addresses the biological mechanisms your model revealed. The program must be practical, evidence-based, and measurable.

Engineering Design Thinking

1. WHAT SPECIFIC STRESS-REDUCTION INTERVENTIONS DOES YOUR PROGRAM INCLUDE AND WHICH BIOLOGICAL MECHANISM DOES EACH ONE TARGET?

2. HOW DOES YOUR PROGRAM ADDRESS THE SOCIAL SUPPORT COMPONENT THAT YOUR MODEL SHOWS IS CRITICAL FOR CORTISOL BUFFERING?

3. WHAT MODEL EVIDENCE SUPPORTS YOUR CLAIM THAT THESE INTERVENTIONS WILL MEASURABLY IMPROVE STUDENT HEALTH?

4. HOW WOULD YOU MEASURE WHETHER YOUR PROGRAM ACTUALLY REDUCES CORTISOL LEVELS AND IMPROVES IMMUNE FUNCTION OUTCOMES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Psychoneuroimmunologists study the interactions between psychological processes, the nervous system, and the immune system, earning $70,000-$140,000/year in research positions. Health psychologists apply stress-health research in clinical settings, earning $75,000-$120,000/year. Public health specialists designing workplace and school wellness programs earn $55,000-$100,000/year.